# CPE 439 Lecture 01

Comments

Administrative (Syllabus)

Course Overview

# Word of the Day

- **Abstraction**: a way of simplifying a concept by removing some of the details involved.

# CPE 439 Issues

- Spring 2012:
  - the last time I taught the course
  - The last time I worked with hardcore MCUs
- Different world back then
  - MCU was simple (Atmel AVR)
  - IDE was simple
  - Not possible for hardware debugging
  - The assembly language was understandable
  - Every aspect of the C code was understandable

# CPE 439 Issues: The Bad News

- I’m just not to where I want to be with the course, including:
  - The IDE
  - HAL
  - MCU hardware
  - Architecture & interrupt architecture
  - Topic outline
  - Labs/Experiments

# CPE 439 Issues: The Good News

- Embedded systems was always what I wanted to do
- I'm only teaching this one class, so I'm focused on coming up to speed on the material
- The RTOS principles didn't change (but FreeRTOS has more features)
- I'm out to learn a lot…
- We've got the world's best TA

# CPE 439 Final Thoughts

- Not going to be the hardest course
- I'm not out to "get" anyone
- There's going to be days I have nothing to say
- Hopefully supplemented with experiments
- The material is the coolest stuff; any embedded person should be familiar with it
- I do feel you'll take useful information away from the course, though maybe not as much as you and I hope for
- *Let's see how it goes??*

# CPE 316 & EE 329 & STM32CubeIDE

- STM32CubeIDE changed
  - It’s now in two pieces
    - MX: generates code
    - IDE: does everything else
- CPE 316 & EE 329 Instructors: HAL or not?
  - How far did you get?
  - What were some of the projects you did?
  - How familiar were you with the IDE?
  - How familiar were you with the STM32 part?

# CPE 439: Course Structure

- Weekly Quizzes
  - “open source”
- 4-6 Experiments
- RTOS Survey Projects
- Final Project
  - RTOS research or project
- Lab Final (short answer)
- Lecture Final (solved problems)

# Embedded Systems Overview

- Digital Design: Using a digital circuit to solve a problem
  - CPE 133: digital circuit under hardware control
  - CPE 233: digital circuit under low-level program control
  - CPE 316: digital circuit under high-level program control

- What the circuits did: the waited for input and reacted accordingly (with outputs?)

# Embedded Systems Overview

- Embedded System: A specialized computer system that has a dedicated function within a larger mechanical or electronic system
- What an embedded system does: it reacts to input and eventually outputs something
- Monitored inputs generally had a nice level of separation
  - which we will forever refer to as tasks

# Embedded Systems Overview

- CPE 133: One task? *(easy)*
- CPE 233: Two tasks? *(easy)*
  - My CPE 233: I covered interrupts
- CPE 316: Multiple tasks *(not so easy)*

- <u>Embedded Systems:</u> They generally handled multiple tasks.
- More tasks:
  - Complicated system
  - Response time issues

# Embedded Systems Overview

- Embedded systems need to handle multiple tasks but there is only one processor
  - Sounds complicated and problematic
- You had to write an operating system for the circuit to deal with multiple tasks:
  - My definition of an OS: the interface (hardware or software) between the outside world and the hardware.
  - It worked but… things were relatively simple
    - Simplicity help handle time constraints

# Embedded Systems Overview

- Many tasks but one MCU:
  - Deal with complication with some extra software tools: use an RTOS
- Real Time Operating System: an operating system for real-time computing applications that processes data and events that have critically defined time constraints
  - Which means extra layer of code that provides tools to make systems that handle multiple tasks more manageable.

# Embedded Systems Overview

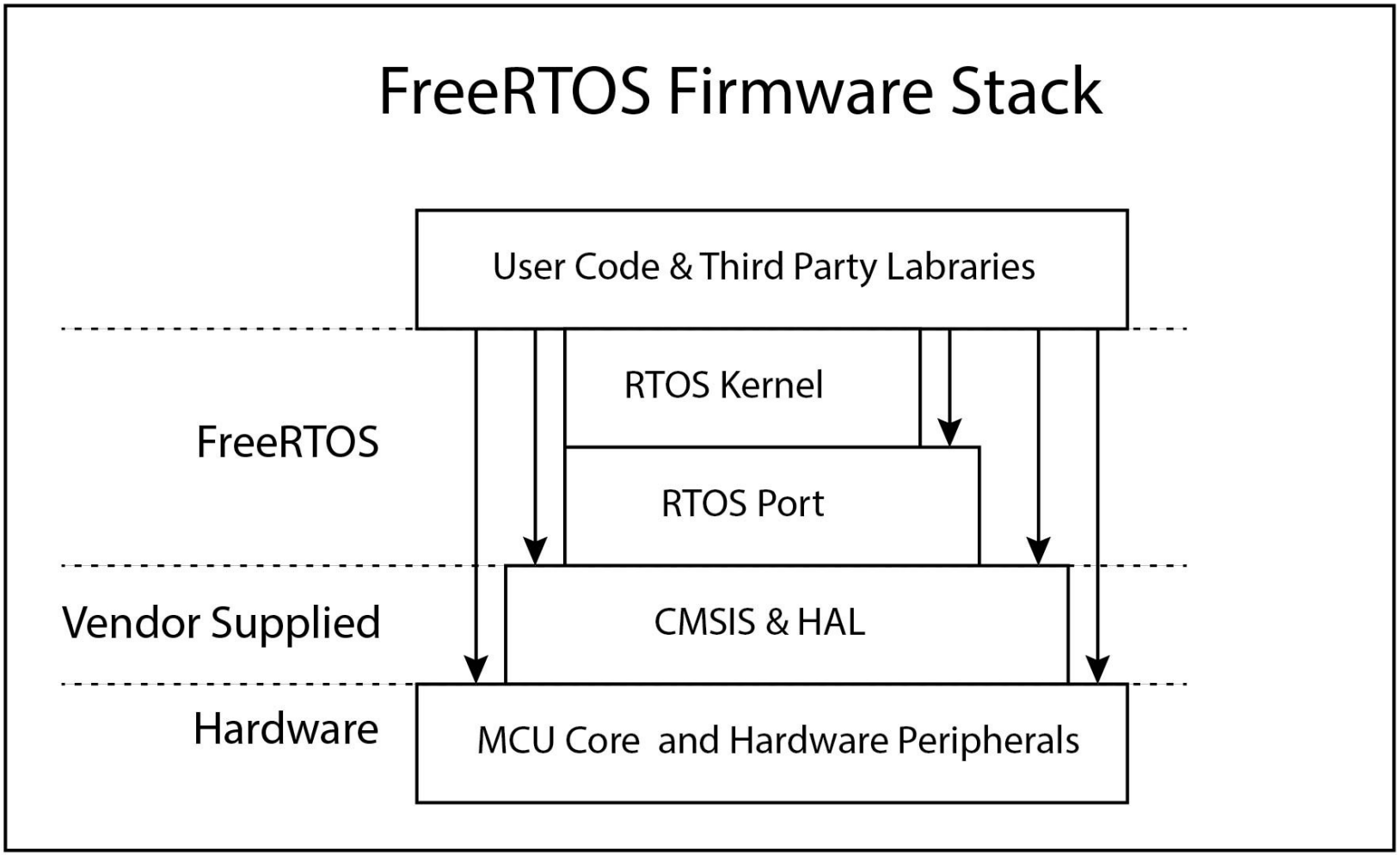
- Recall our embedded system deals with tasks
  - The problem: embedded systems treat all tasks with the same “level of importance”.
    - Processor may get stuck doing something unimportant when something important needs doing.
- RTOS solution:
  - Assign priorities to tasks
  - Ensure MCU is always processing highest priority task

# Embedded Systems Overview

- Event detection & handling
  - Interrupts vs. Polling
    - Aren’t interrupt higher priority?
- Other issues
  - Process events inside ISRs?
    - ISRs processing fast but are other events being ignored?
  - Process event outside of ISRs?
    - How? Which event to process first?

# Embedded Systems Overview: The RTOS:

- Provides tools to process events in something that appears to be real time
  - Quick responses to important events
  - Tools are generally function calls to the RTOS
  - Gives appearance of multitasking

- Your mission in CPE 439:
  - Understand the problems and solutions (tools)
  - Use the tools to design working systems

# Embedded Systems Overview